

INTRODUCTION

This project is an automatic light-controlled switching system that uses analog electronics to operate a load based on ambient light conditions. The core idea is to sense changes in light intensity using Light Dependent Resistors (LDRs) and convert those changes into corresponding electrical signals. These signals are then processed using operational amplifiers configured as comparators, which continuously compare the sensed voltage with a predefined reference level.

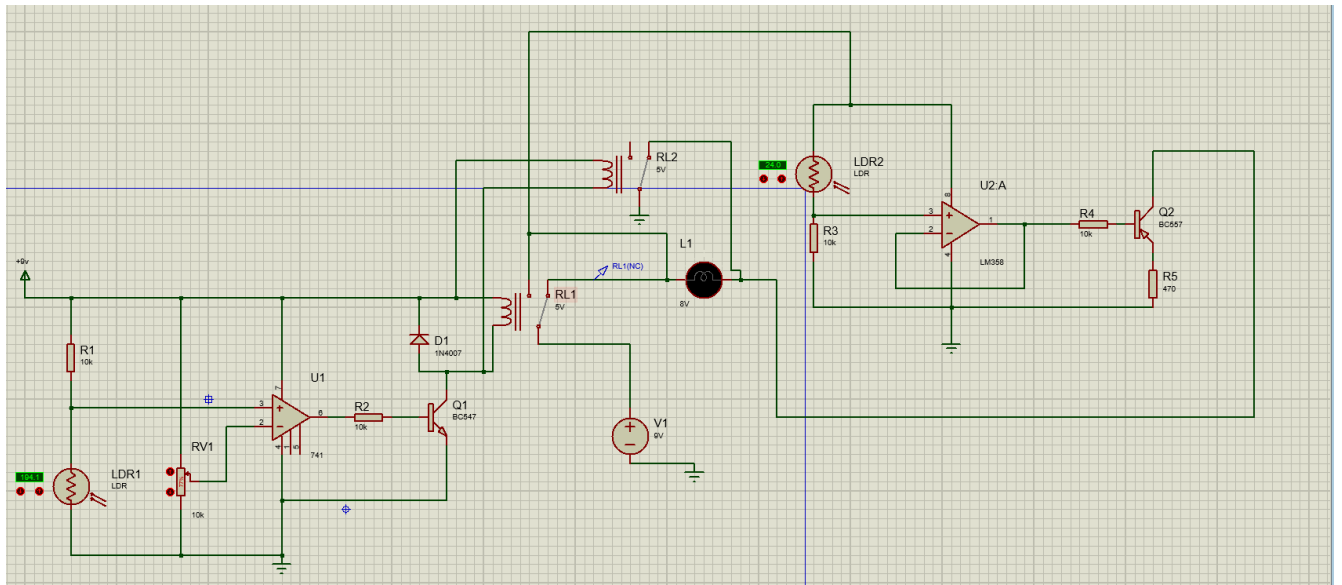
When the light intensity crosses the set threshold, the comparator output changes state, driving a transistor that acts as a switching interface. The transistor amplifies the current and energizes a relay, which in turn controls the power supplied to the load, such as a bulb. In this way, the system automatically turns the load ON or OFF depending on environmental lighting conditions without manual intervention.

Overall, the project demonstrates the practical integration of sensing elements, analog signal processing, and switching devices to build an efficient and autonomous control system, commonly used in applications like street lighting and energy-saving automation.

AIM

- To design a Night Lamp for Residential Lighting Applications
- To detect ambient light levels using a light sensor.
- To reduce power consumption and improve energy efficiency.

CIRCUIT DIAGRAM



CIRCUIT LEVEL DESCRIPTION

At the circuit level, both stages implement the same signal chain: a resistive voltage divider converts a resistance change into a voltage, an op-amp comparator makes a binary switching decision, and a transistor translates that decision into current flow through the load. The following describes each stage in detail.

Stage 1 — LDR1 / 741 Op-Amp / Relay Control

Voltage Divider & Signal Generation

LDR1 and R1 are connected in series between the supply rail and ground, forming a resistive voltage divider. Because LDR resistance decreases with increasing light intensity, the voltage at the LDR1–R1 junction rises with brightness and falls with darkness, producing a direct analog representation of ambient light. This voltage is applied to one input of the 741 op-amp.

Comparator Operation

Potentiometer RV1 is wired as a second voltage divider across the supply, providing an adjustable reference voltage to the other input of the 741. The op-amp operates in open-loop (comparator) mode. In this configuration, even a small differential between its two inputs drives the output to saturation, either near the positive supply rail or near ground, generating a clean binary output.

Transistor Switching — Q1 (NPN)

The op-amp output is applied to the base of NPN transistor Q1 via current-limiting resistor R2. When the output goes high, the base-emitter junction of Q1 is forward-biased (≥ 0.7 V), driving Q1 into saturation. In saturation, Q1 presents a near short-circuit between collector and emitter, connecting one terminal of relay coil RL1 to ground. The opposite coil terminal is tied to the supply, so current flows through the coil, energizing the relay.

Relay & Load Switching (RL1, L1)

The relay provides galvanic isolation between the low-power control circuit and the higher-power lamp circuit. When energized, its contacts change state, completing the circuit that powers lamp L1 from the supply. When Q1 turns off, the relay de-energizes and the lamp circuit is broken.

Flyback Protection — D1

Diode D1 is connected in reverse bias across relay coil RL1. During normal operation, it is non-conductive. When Q1 turns off, and the magnetic field in the coil collapses, the resulting inductive voltage spike is clamped by D1, safely dissipating the energy and protecting Q1 from breakdown.

Stage 2 — LDR2 / LM358 Op-Amp / PNP Transistor Control

Voltage Divider & Signal Generation

LDR2 and R3 form an identical resistive voltage divider, producing a light-dependent voltage that is applied to one input of the LM358 op-amp. Unlike the 741, the LM358 is designed for single-supply operation, allowing its output to swing closer to ground and operate reliably at lower supply voltages.

Transistor Switching — Q2 (PNP, BC557)

The LM358 output is fed to the base of PNP transistor Q2 (BC557) via resistor R4. As a PNP device, Q2 requires its base to be driven below the emitter potential to conduct. When the op-amp output pulls low, the base-emitter junction becomes forward-biased and Q2 turns on, allowing current to flow from emitter to collector through R5 and the connected load. When the op-amp output goes high, the base approaches emitter potential, turning Q2 off. This stage provides an independent, solid-state auxiliary switching output without the need for a relay.

WORKING PRINCIPLE

The circuit functions as an automatic, ambient light-controlled lamp switching system. It continuously monitors environmental light levels and toggles the lamp on or off in response, without any manual intervention. The system is divided into two independent sensing and control stages, each responsible for a distinct switching function.

Stage 1 — Primary Light Sensing & Lamp Control

A light sensor detects ambient brightness and feeds a proportional voltage signal to a comparator. The comparator continuously evaluates this signal against a user-defined threshold, adjustable via a potentiometer. When the ambient light drops below the threshold, indicating darkness, the comparator triggers a transistor, which in turn energises a relay. The relay contacts close, connecting the lamp to the power supply and turning it on.

When light rises back above the threshold, the comparator reverses its decision, the transistor switches off, the relay de-energizes, and the lamp turns off. A protection diode prevents the voltage spike generated during relay switch-off from damaging the transistor.

Stage 2 — Secondary Sensing & Auxiliary Control

A second, independent light sensor feeds a separate comparator. Rather than controlling a relay, this stage drives a transistor directly, providing an auxiliary switching output. This enables a second configurable light threshold, allowing the system to refine its sensitivity, introduce staged responses, or conditionally enable a separate branch of the circuit depending on lighting conditions.

Both stages follow the same fundamental workflow:

Sense Light → Convert to Voltage → Compare Against Threshold → Switch Load

The outcome is a fully autonomous system that reacts to real-world light conditions and controls the lamp without operator input.

DESIGN IN DETAIL

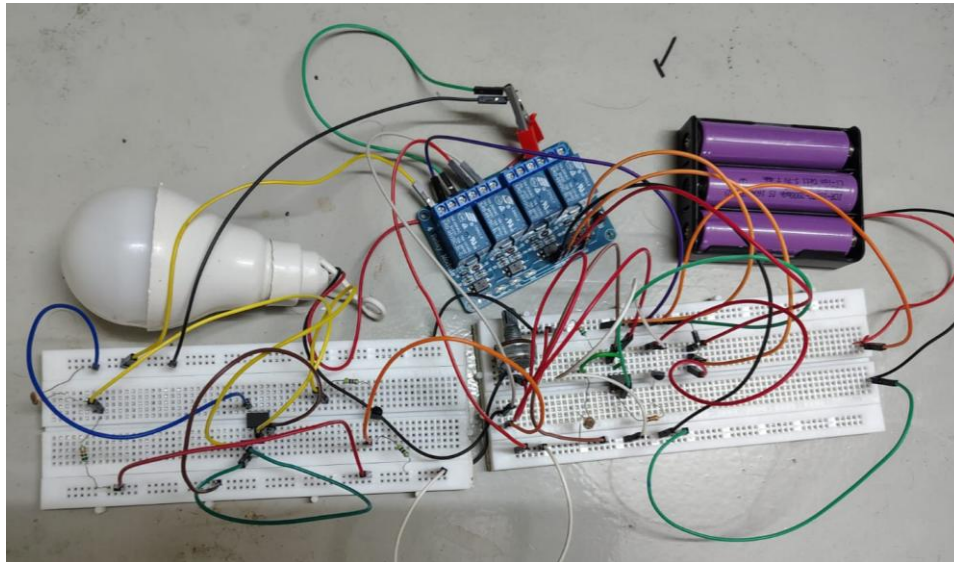
COMPONENT	SPECIFICATION
R1,R2,R3,R4	10K Ω
R5	470 Ω
U1	LM741OPAMP
U2	LM358 OP AMP
LDR(2)	-
Q1,Q2	BC547, BC557 TRANSISTOR
D1	1N4007
Power supply	12V
Bulb	12 volt Rating
Relay	Nominal Voltage = 5 volt Internal Resistance = 70 Ω

OUTPUT (BASED ON THE CONDITION)

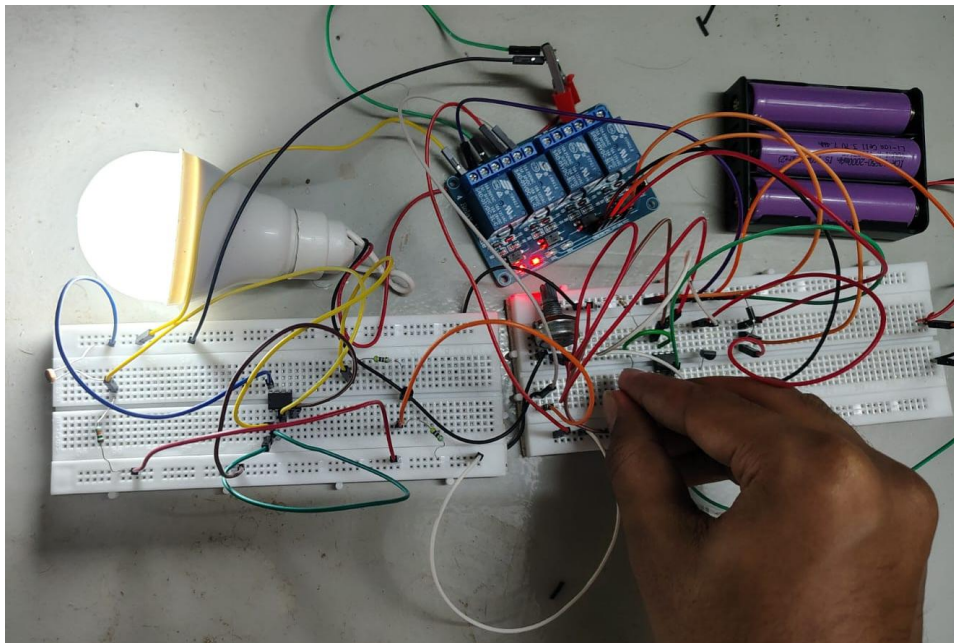
CONDITION	OUTPUT
DAYLIGHT	LIGHT OFF
DARKNESS	LIGHT ON
DIM LIGHT	LIGHT ON PARTIALLY

HARDWARE IMPLEMENTATION OF THE CIRCUIT

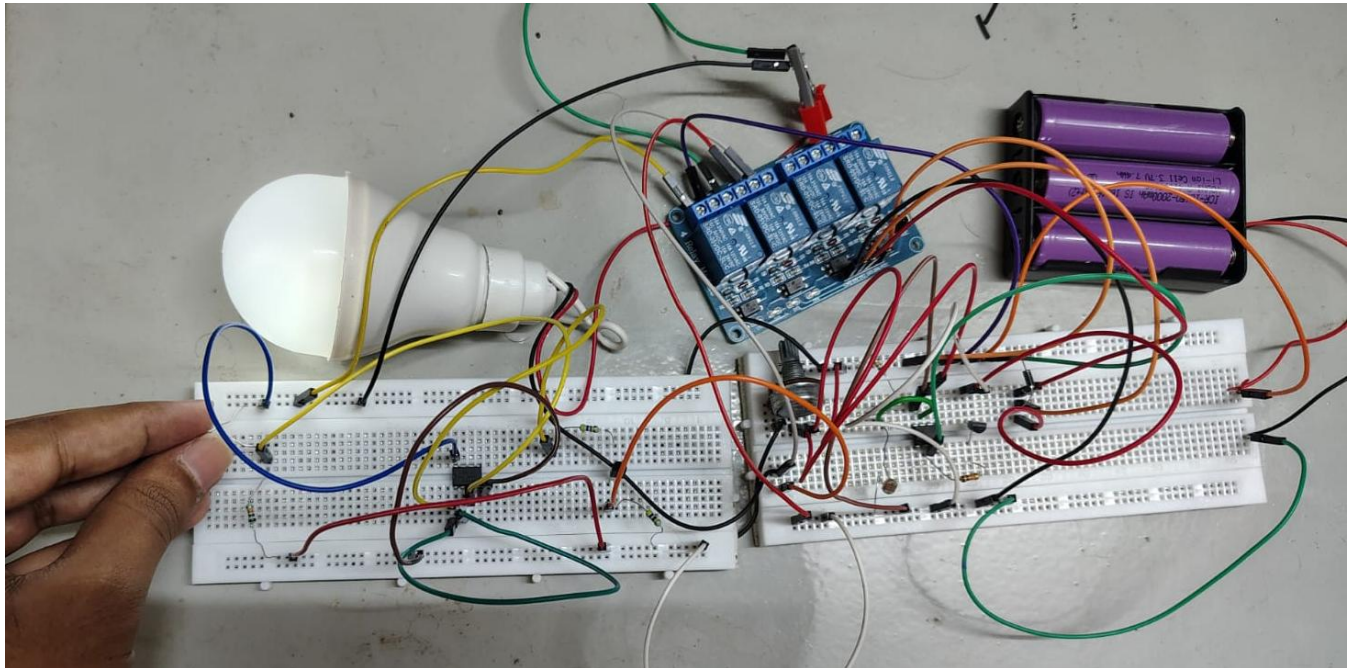
CASE I: In presence of light



CASE II: In complete darkness



CASE III: In optimal darkness



INFERENCE

- The system automatically controls light based on ambient light intensity.
- During the daytime, the light remains OFF.
- During the night, the light turns ON automatically.
- The system reduces manual work and power consumption.
- It improves energy efficiency and automation.

CONCLUSION

This project shows how an LDR (Light Dependent Resistor), LM741 Operational Amplifier, and BC547 Transistor can be used together to make an automatic light-sensing circuit. The circuit works by detecting the amount of light and switching the LED ON or OFF automatically. This makes the system simple, useful, and efficient.

The main importance of this project is that it introduces the concept of automation and sensor-based control in electronics. Its advantages are low cost, easy circuit design, less power consumption, and reliable operation. Such circuits are necessary in daily life applications like automatic lights, emergency lighting, and security systems, where manual switching is not practical. Overall, this project is a simple and effective way to understand practical electronic automation.